

Welcome and Logistics

POLI3148 Data Science in Politics and Public Administration

Prof. Chen Haohan

This lecture

- Some General Introduction
- The Plan
- Logistics

About Me



Prof. CHEN Haohan 陳昊瀚

- A data scientist
- A comparative political scientist
- A coffee lover
- A cat person



What I do for My Research and Teaching

Apply computational methods to media “big data” to understand:

- Elites and the public’s strategic political communication
- Ideology and identity

What I do for My Research and Teaching

Media platforms am studying...

Some conventional outlets



豆瓣 *douban*

CCTV



知乎



What I do for My Research and Teaching

A few somewhat less well-understood outlets...



Teaching Assistant: Du Baichen



- DU Baichen 杜柏辰
- BSocSc and MA in Linguistics@HKU
- Majored in PPA and Linguistics
- Was in POLI3148 in 2021

Teaching Assistant: Liao Li



- INFP
- Otaku and data scientist (self-proclaimed)
- Study bureaucratic politics and authoritarian politics
- Fan of Kazuo Ishiguro

The Plan

Why Data Science and POLI3148?

Data science is extraordinary



Increasing capacity of of big data and AI



Why Data Science and POLI3148?

Data science is ordinary

- *Typists* in the 1990s = Data scientists now
- Soon-to-be part of essential professional skills

Debunking two myths about data science

Myth: Social sciences and humanity trainees/ professionals are losing jobs to, technicians (computer scientists, data scientists...), AI

- Yes and no
 - **Yes:** mechanical human jobs are being replaced
 - **No:** Deep understanding of the society matters; People will be the final decision-maker (for now)

Myth: You need to be good at maths/ technologies to learn data science.

- Yes and no
 - **Yes:** learning the “why” needs background knowledge
 - **No:** learning the “how” is getting easier -- consider it learning a language

A walkthrough of lecture topics

Date	Week	Lecture	Readings	Due
09/03	1	Welcome & Logistics		
09/10	2	Python Basics (1)	Kaggle: Intro to Programming	
09/17	3	Python Basics (2)	Kaggle: Intro to Python	P1
09/24	4	Data Wrangling	Kaggle: Pandas & Data Cleaning; V-Dem	P2
10/01	5	<i>Public holiday. No class.</i>		
10/08	6	Data Visualization (1)	Kaggle: Data Visualization; V-Dem	P3
10/15	7	<i>Reading Week. No class.</i>		
10/22	8	Data Visualization (2)	Kaggle: Geospatial Analysis; UCDP	P4
10/29	9	Text Analysis (1)	Hovy p. 1-49; CMFA	A1
11/05	10	Text Analysis (2)	Hovy p. 50-83; CMFA	B1
11/12	11	Machine Learning (1)	Kaggle: Introduction to Machine Learning; Lee&Shih	P5
11/19	12	Machine Learning (2)	Kaggle: Introduction to Deep Learning; Lee&Shih	
11/26	13	Advanced topic or catch up	none	

The Plan: Become a Data Scientist in 90 Days!

- Python Basics
- Data Wrangling
- Data Visualization
- Text Analysis
- Machine Learning

Python Basics

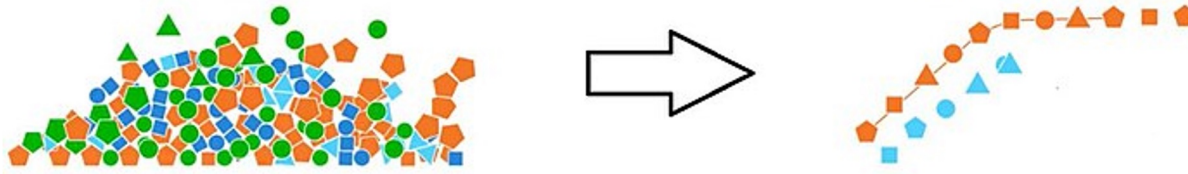
“The programming language in the AI era.”

- Getting started with programming
- Basic Python syntax
- Python commands relevant to data science



Data Wrangling

Load, clean, and transform data for analysis



Starting data

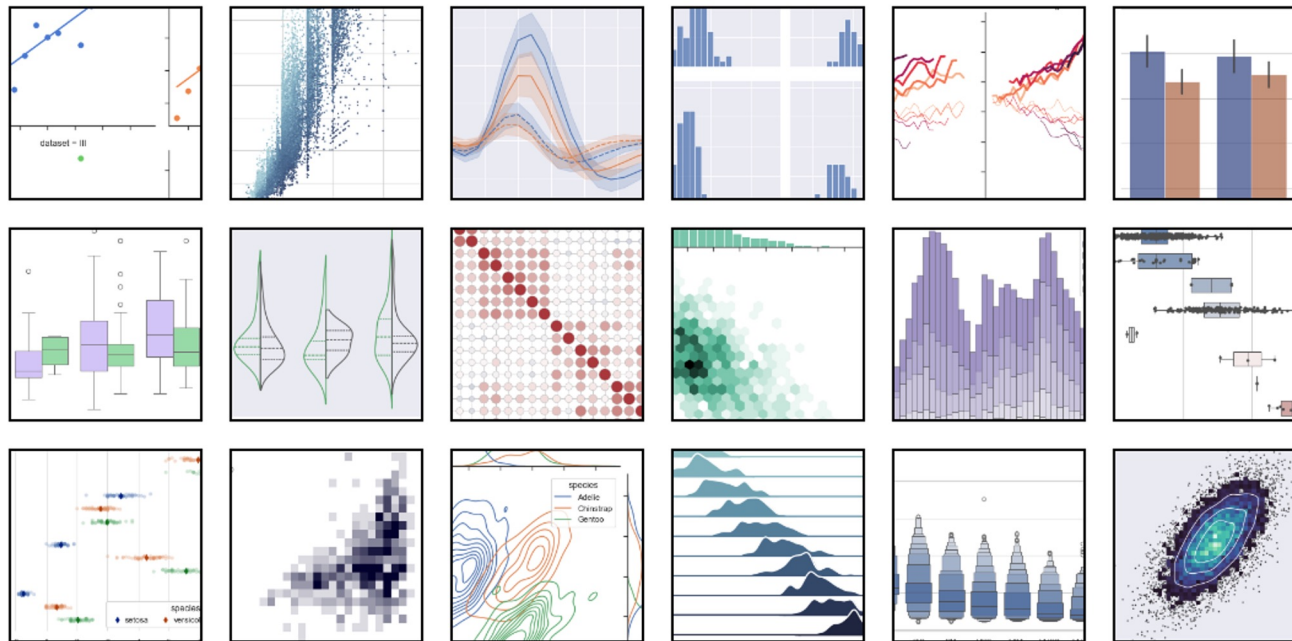
Name	Phone	Birth date	State
John, Smith	445-881-4478	August 12, 1989	Maine
Jennifer Tal	+1-189-456-4513	11/12/1965	Tx
Gates, Bill	(876)546-8165	June 15, 72	Kansas
Alan Fitch	5493156648	2-6-1985	Oh
Jacob Alan	156-4896	January 3	Alabama

Result

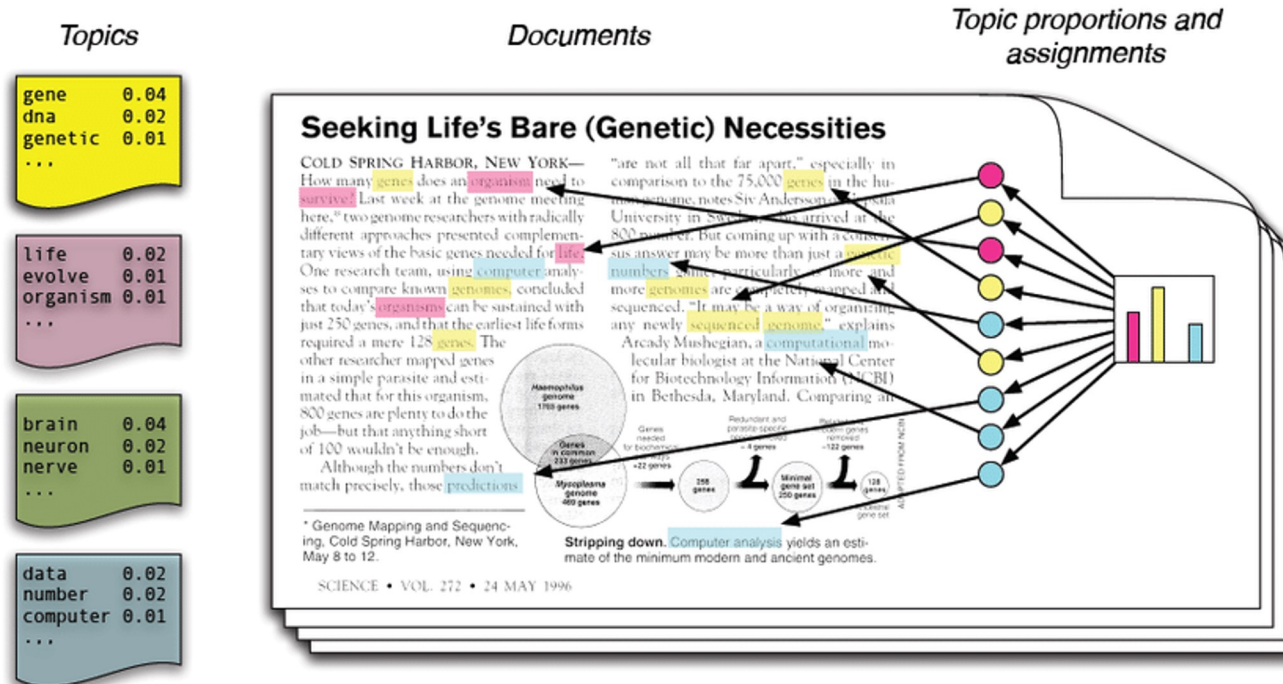
Name	Phone	Birth date	State
John Smith	445-881-4478	1989-08-12	Maine
Jennifer Tal	189-456-4513	1965-11-12	Texas
Bill Gates	876-546-8165	1972-06-15	Kansas
Alan Fitch	549-315-6648	1985-02-06	Ohio

Data Visualization

Turn numbers to graphics

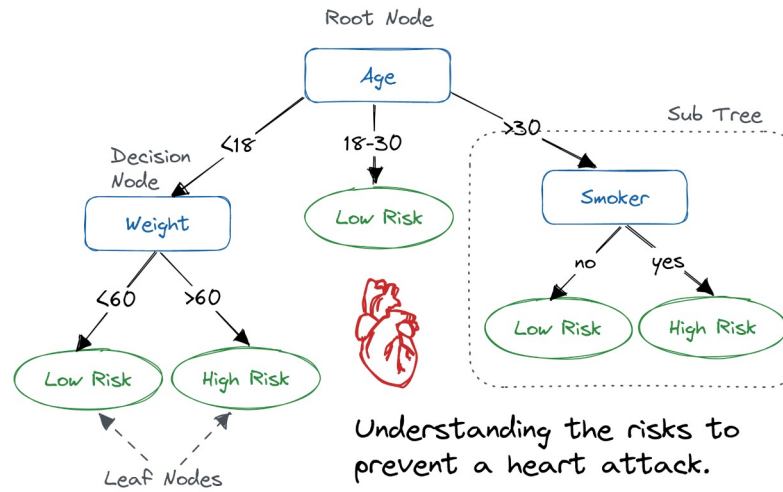
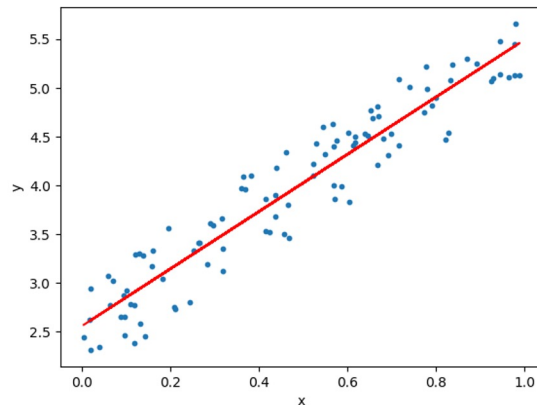


Text Analysis



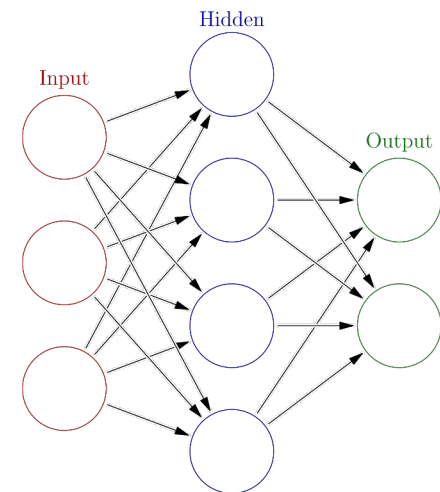
Machine Learning

Linear regression



Decision tree

Neural network



So many things to learn. So little time we have...

Design of teaching materials

- Very selective coverage
- Interactive reading materials
- Case studies

Selective Coverage: a “Dim Sum” Approach

Breadth > Depth. Samples a few topics to help with basic understanding



Interactive reading materials: Read and practice

Read differently from your other courses

- Short, focused online modules with interactive practicing opportunities
- My objective is to teach you “how to fish” 
 - Learning what we cover
 - Learning how and where to learn what we do not covered

Learn from doing

- Pick A few datasets I love
 - Global politics
 - International relations
 - Chinese politics
- Apply related data science skills

Wrangling and visualizing global socio-economic data

v-dem.net

A rich cross-national datasets with

- Economic and political indicators
- In the past hundreds of years!

Tracking global conflict data with geo-referenced data

ucdp.uu.se

Geo-coded data of organized *violence* (phenomena of lethal violence occurring at a given time and place).



Understanding China's diplomatic strategies with text analysis

Mochtak, Michal, and Richard Q. Turcsanyi. "Studying Chinese foreign policy narratives: Introducing the Ministry of Foreign Affairs press conferences corpus." *Journal of Chinese Political Science* 26, no. 4 (2021): 743-761.

Text transcripts of press conferences of China's Ministry of Foreign Affairs

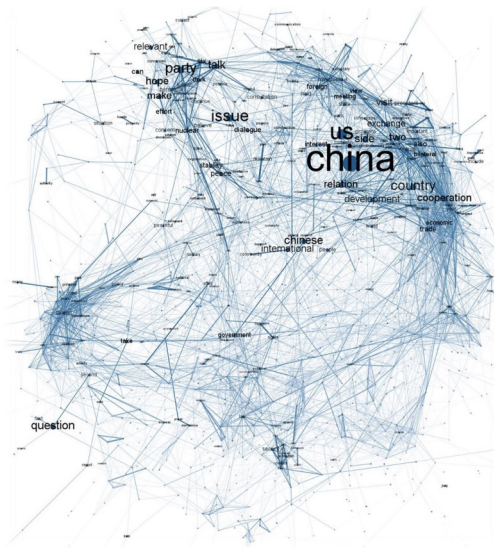


Fig. 5 Semantic network tracing US-China discourse under President Hu Jintao



Demystifying Chinese Elite Politics with Machine Learning

Creative use of machine learning for political science research

- Loyalty or competence? A debate in the study of Chinese politics
- Answer with
 - A Biographical dataset of Chinese officials
 - A machine learning approach

Paper and data:

- <https://doi.org/10.1073/pnas.230514312>
- <https://chinadatalab.ucsd.edu/elites/>

Birthplaces of Chinese political elites visualized



Objective of these Case Studies

Develop your “computational thinking” capacity

- Apply computational methods to political data
- Give meaningful interpretation to outputs of computational methods

Assessment and Feedback

Output and Assessment

Type	Output	Weight	Due
Participation (30%)	[P0] Lecture Participation	10%	-
	[P1-5] 5 In-Class Exercises. Each weighs 4%.	20%	-
Assignments (40%)	[A1] Problem Set 1: Data Wrangling & Data Visualization	20%	10/29
	[A2] Problem Set 2: Text Analysis and Machine Learning	20%	12/17
Group Final Project (30%)	[G1] Poster Presentation at DaSPPA Festival	20%	12/07
	[G2] Project Portfolio	10%	12/14
Bonus (10% max.)	[B1] Project Proposal	5%	10/29
	[B2] Others	5%	-

Lecture Participation 10%

- Ask questions.
- Ask me to clarify the lecture stop making sense.
- No. Don't "figure it out later."

Btw: Make yourself comfortable in class!

Food, snacks, and drinks are perfectly fine (no strong odor or sound though)

In-class exercises 20%

- Learn data science = learning a second language
- You need constant practice (not binge working in the end)
- Perform data analysis under close guidance
- About 5 in-class exercises related to
 - Readings
 - Case studies
- How
 - Get started in class
 - Do some extension after class and submit a few days later

Problem Sets 20% X 2

- Case studies
- Perform data analysis under medium level of guidance
- Mixture of technical questions and open-ended questions
- Assigned ~2 weeks before due

Group Final Project 30% + bonus 5%

- Open-ended task
- Find your favorite data and methods and do creative work with them!
- Outputs
 - Project proposal + bonus 5%
 - DaSPPA Festival Poster Presentation 20% (**Dec 3**)
 - Project Portfolio 10%

Note: You absolutely need to attend **the poster presentation** in person to claim the 20%. No exception.

Project topics from last year

- China's rice export
- Democracy and press freedom
- Public attitudes towards the Israel-Palestine Conflict
- CCP's changing policy focus
- Collective memory of the Cultural Revolution
- China's online philanthropy
- Global energy policy design

Grading

- Reward good work. Also reward efforts and progress
- Nicely Fair & Fairly Nice

Speak with me

- Office hours on Wednesdays 2 - 5pm.
- I can do both in-person and online meetings
- Make appointments through Calendly
<https://calendly.com/haohanchen/office-hours>

Logistics

To start with...

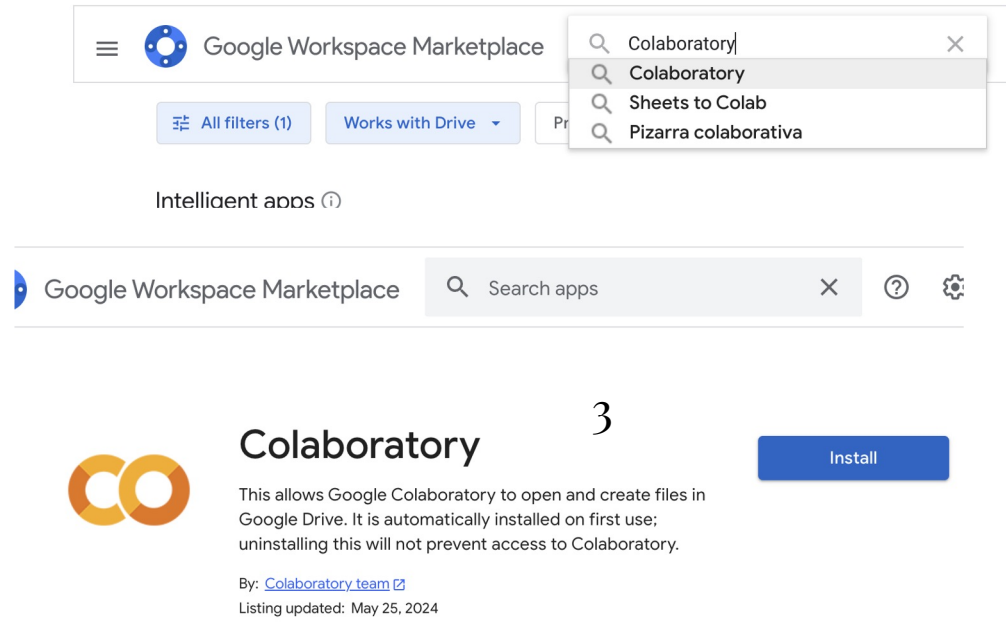
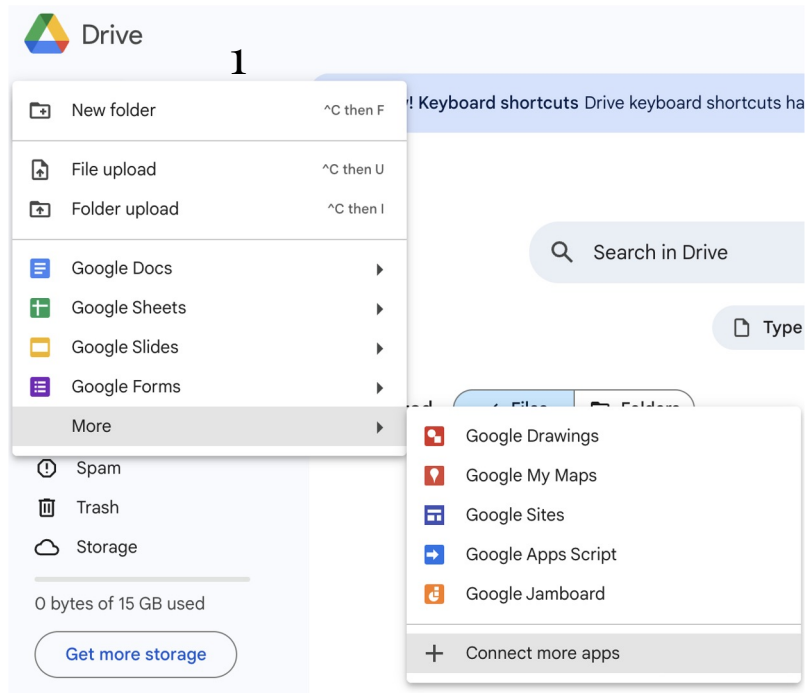
- We are going to use Google's amazing data science toolkit.
- You need a Google Account (free of charge for registration)

Sign up a Google Account at google.com if you don't have one.

Cloud Computing Platform: Google Colab

Open Google Drive. Setup Google Colab on your Google Drive.

2



3

Feedback System: Google Classroom

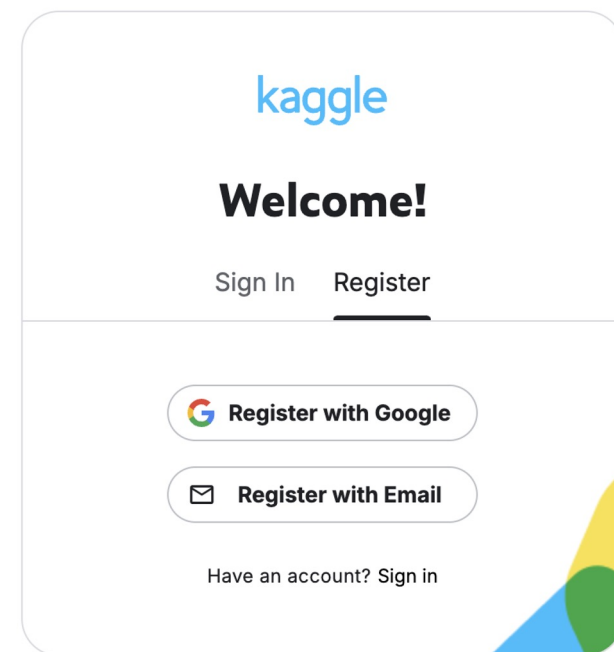
Join our course's Google Classroom

- Use invitation link on Moodle
- Fill the survey that helps us associate you with your Google Account

Setup your interactive “textbook”

Interactive Readings and exercise: Kaggle

<https://www.kaggle.com/>



Additional information about the logistics

Refer to Google Classroom